

The Role of Expectations in Monetary Policy

Chapter 24

Econ 351 – Monetary Economics

University of Tennessee, Knoxville

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Preview

This chapter examines how **expectations** — particularly **rational expectations** — have transformed our understanding of monetary policy.

Central questions we will address:

1. Why can't policymakers rely on **historical relationships** to predict the effects of new policies?
2. Should central banks follow **rules** or exercise **discretion**?
3. Why is **credibility** the most valuable asset a central bank can possess?
4. How can central banks **build and maintain** credibility over time?

Outline

The Lucas Critique

Policy Rules vs. Discretion

Credibility and the Nominal Anchor

Credibility and Anti-Inflation Policy

Maintaining Credibility

The Lucas Critique of Policy Evaluation

Background

Robert Lucas (1976) challenged the way economists had been evaluating policy for decades.

The Traditional Approach

- ▶ Policymakers simulate new policies using **macroeconometric models**
- ▶ These models are estimated from **historical data**
- ▶ They assume people will react **the same way as in the past**

Lucas's Insight

When policy changes, people **change their expectations and behavior**. Therefore:

- ▶ Relationships estimated from past data may **no longer hold**
- ▶ Policy evaluation based only on historical correlations can be **misleading**

Why Expectations Matter for Policy

The Mechanism

- ▶ If people expect **higher inflation**, they demand higher wages $\Rightarrow$ firms raise prices $\Rightarrow$ inflation actually rises
- ▶ If people expect **higher interest rates**, they adjust saving and investment decisions *today*
- ▶ These behavioral changes **alter the very relationships** that the model was built on

Why Expectations Matter for Policy

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The Bottom Line

Any model that treats expectations as **fixed** while policy changes will produce **unreliable forecasts**. Good policy analysis must account for how expectations respond to the policy itself.

Application: The Term Structure of Interest Rates

The Expectations Hypothesis

Long-term interest rates reflect the market's expectations of **future short-term rates**:

$$i_n = \frac{1}{n} (i_1 + i_1^e + i_2^e + \cdots + i_{n-1}^e) + \ell_n$$

where ℓ_n is the term (liquidity) premium.

Temporary Rate Increase

- ▶ Short-term rates rise briefly
- ▶ Expected future rates unchanged
- ▶ Long-term rates change **only slightly**

Permanent Rate Increase

- ▶ Short-term rates rise
- ▶ Expected future rates **also rise**
- ▶ Long-term rates rise **significantly**

Lucas Critique Applied to the Term Structure

What Traditional Models Predict

- ▶ Based on historical data, a short-term rate increase has only a **small effect** on long-term rates
- ▶ The model assumes people respond the **same way as before**

What the Lucas Critique Says

- ▶ If the policy **regime changes** (e.g., the Fed commits to permanently higher rates), expectations adjust
- ▶ Long-term rates may rise **much more** than the historical model predicts
- ▶ The model's forecast becomes **inaccurate**

Lesson

Policymakers who ignore expectation effects may **underestimate** or **overestimate** the true impact of their actions.

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The Fundamental Question

Should monetary policy follow **fixed rules** or allow **policymaker discretion**?

Rules

- ▶ Policy is set **in advance** according to a formula or commitment
- ▶ Limits policymakers' flexibility
- ▶ Provides **predictability** to markets

Discretion

- ▶ Policymakers **adjust policy as needed** based on current conditions
- ▶ Allows creative responses to unforeseen events
- ▶ Requires **trust** in policymakers' judgment

The Time-Inconsistency Problem

The Temptation

1. Central bank **promises** low inflation $\Rightarrow$ people set low π^e
2. Central bank then **surprises** with expansionary policy $\Rightarrow$ short-run output gains
3. But people **catch on** $\Rightarrow \pi^e$ rises $\Rightarrow$ inflation rises with **no lasting output gain**

The Result

- ▶ Short run: lower unemployment and higher output (temporarily)
- ▶ Long run: **higher inflation, no output gain, and loss of credibility**
- ▶ Once credibility is lost, it is **very costly to rebuild**

Time Inconsistency: Kydland and Prescott



Finn Kydland, lecturing in 2015.

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Edward Prescott, 2015.

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Kydland and Prescott (Nobel 2004) formalized the **time-inconsistency problem**: a policy-maker free to re-optimize each period is tempted to renege on a low-inflation promise, so under discretion the public expects higher inflation — and gets it, with no lasting output gain.

Types of Policy Rules

Nonactivist Rule: Constant Money Growth (Friedman)

- ▶ Central bank increases the money supply at a **fixed rate** (e.g., 3% per year)
- ▶ Does **not respond** to short-run economic fluctuations

Advantages:

- ▶ Prevents inflationary mistakes
- ▶ Reduces political pressure

Criticisms:

- ▶ Cannot respond to recessions
- ▶ Ignores changes in money velocity

A **nonactivist rule** sets policy in advance and does **not respond** to the state of the economy; an **activist rule** (next slide) adjusts the policy instrument to the inflation and output gaps, but still by a fixed, announced formula.

Activist Rules: The Taylor Rule

A more flexible rule that adjusts interest rates based on economic conditions:

$$i = \pi + r^* + a_{\pi}(\pi - \pi^T) + a_Y(Y - Y^P)$$

- ▶ r^* : equilibrium real interest rate
- ▶ $a_{\pi} > 0$: response to the **inflation gap**
- ▶ $a_Y > 0$: response to the **output gap**

Advantages:

- ▶ Responds to economic conditions
- ▶ Transparent and predictable
- ▶ Provides a benchmark for policy

Criticisms:

- ▶ Requires accurate estimates of Y^P and r^*
- ▶ May be too rigid during crises

The Case for Rules

Why Rules Can Be Superior

- ▶ Avoid the **inflationary bias** created by the time-inconsistency problem
- ▶ Increase central bank **credibility** and **transparency**
- ▶ Reduce **political influence** on monetary policy

Historical Example: The Political Business Cycle

Before the **1972 U.S. presidential election**, President Nixon pressured Fed Chair Arthur Burns to keep interest rates low to boost the economy.

Result: short-term economic boost followed by **high inflation throughout the 1970s**.

⇒ A rule-based framework could have prevented this political manipulation.

Why Discretion Can Be Necessary

- ▶ The economy is complex — rigid rules may become **outdated** as the economy evolves
- ▶ Unforeseen crises require **creative, rapid responses**
- ▶ Monetary policy requires **judgment** that no formula can fully capture

Example: The 2008 Financial Crisis

During the crisis, the Fed:

- ▶ Cut interest rates to **near zero** (faster than any rule prescribed)
- ▶ Created **emergency lending facilities** for financial institutions
- ▶ Launched **quantitative easing** — purchasing trillions in assets

A strict rule like constant money growth would have **prevented** these emergency actions.

Constrained Discretion: The Best of Both Worlds

Proposed by Bernanke & Mishkin

Policymakers operate within a **clear framework** (e.g., an inflation target) but retain **limited flexibility** to respond to shocks.

How It Works in Practice: Inflation Targeting

- ▶ Central bank announces a **numerical inflation target** (e.g., 2%)
- ▶ Policy decisions are **transparent** and explained publicly
- ▶ The central bank has **flexibility** in how quickly it returns inflation to target after a shock

Why It Works

- ▶ Combines the **credibility** benefits of rules
- ▶ Maintains the **flexibility** benefits of discretion
- ▶ Adopted by many central banks: Bank of England, ECB, Bank of Canada, Reserve Bank of New Zealand, etc.

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Credibility and Anti-Inflation Policy

Maintaining Credibility

What Is a Nominal Anchor?

Definition

A **nominal anchor** is a monetary policy target that ties down the price level or inflation rate, giving the public a clear reference point for forming expectations.

Three Types of Nominal Anchors

1. **Inflation Targeting** — maintaining inflation around a publicly announced target (e.g., 2%)
2. **Exchange Rate Peg** — fixing the domestic currency to a stable foreign currency (e.g., U.S. dollar)
3. **Money Growth Rule** — increasing the money supply at a fixed, pre-announced rate

Why It Matters

A credible nominal anchor **stabilizes inflation expectations**, which in turn makes monetary policy **more effective** and **less costly**.

Benefits of a Credible Nominal Anchor

1. Prevents Time-Inconsistency

- ▶ A public commitment limits the temptation to pursue short-term stimulus
- ▶ The central bank is **held accountable** to its stated target

2. Anchors Inflation Expectations

- ▶ If people believe the central bank will stick to its target, π^e remains **stable**
- ▶ Stable expectations reduce **inflation volatility** and make shocks easier to manage

3. Enhances Policy Effectiveness

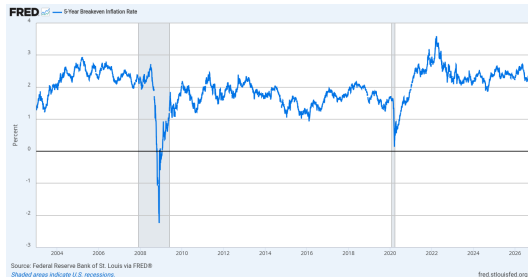
- ▶ With anchored expectations, the central bank can achieve its goals with **smaller policy interventions**
- ▶ Less disruption to financial markets and the real economy

Anchored Inflation Expectations in the Data



Households' expected inflation over the next year (Michigan survey), 1978–2025: from double digits in 1980 to about 3% for three decades.

Source: FRED, Federal Reserve Bank of St. Louis



Market-based 5-year breakeven inflation rate, 2003–2025: expectations stayed near 2% through 2008 and 2020.

Source: FRED, Federal Reserve Bank of St. Louis

Stable π^e is what a credible nominal anchor buys: with expectations anchored, shocks move inflation less and require smaller policy responses.

The Expectations-Augmented Phillips Curve

Inflation Equation

$$\pi = \pi^e + \gamma(Y - Y^P) + \rho$$

- ▶ π : actual inflation
- ▶ π^e : expected inflation
- ▶ $\gamma(Y - Y^P)$: effect of the output gap
- ▶ ρ : supply shock

Key Insight

Inflation depends not only on economic activity ($Y - Y^P$), but **critically** on inflation expectations (π^e).

⇒ If the central bank can **anchor** π^e , it gains enormous leverage over actual inflation.

Credibility and Demand Shocks

With Credibility

After a positive demand shock:

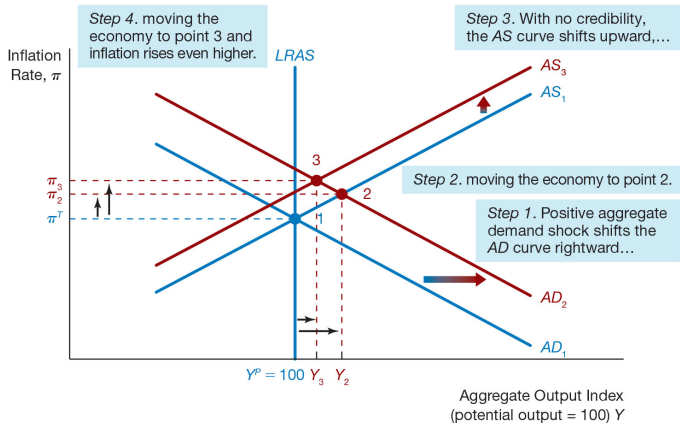
- ▶ AD shifts right, π rises temporarily
- ▶ But π^e remains **anchored** at target
- ▶ AS does **not shift**
- ▶ Tightening policy brings π back to target with **minimal output cost**

Without Credibility

After the same shock:

- ▶ People expect **higher inflation**
- ▶ π^e rises $\Rightarrow$ AS shifts **upward**
- ▶ Inflation rises **even more**
- ▶ Restoring price stability requires a **larger, costlier** policy response

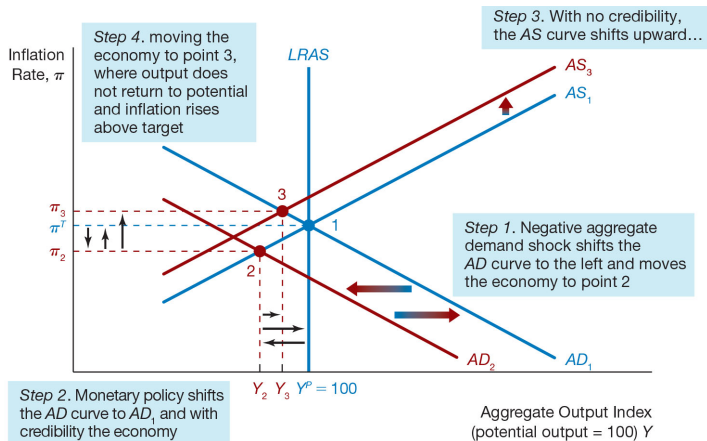
Credibility and Aggregate Demand Shocks (1/2)



With credibility, the economy returns to point 1 quickly.

Mishkin, The Economics of Money, Banking, and Financial Markets

Credibility and Aggregate Demand Shocks (2/2)



Without credibility, AS shifts up and the economy moves to point 3 with higher inflation.

Mishkin, The Economics of Money, Banking, and Financial Markets

Credibility and Supply Shocks

With High Credibility

After a negative supply shock:

- ▶ AS shifts to AS_2 (small shift)
- ▶ π^e remains **anchored**
- ▶ Economy moves to point 2
- ▶ **Smaller** decline in output
- ▶ **Smaller** rise in inflation

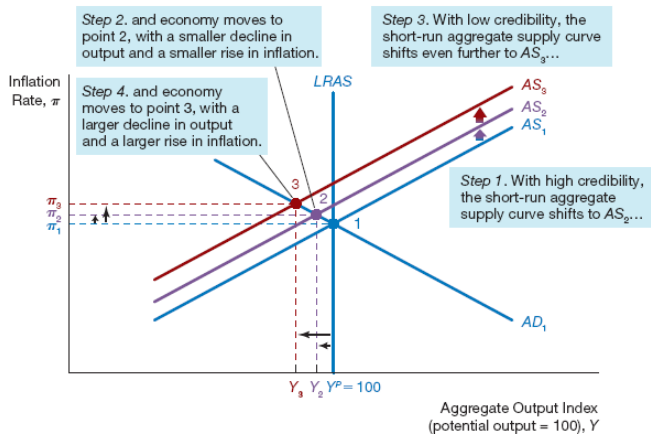
⇒ Credibility acts as a **shock absorber**: the same shock causes less damage when expectations are anchored.

With Low Credibility

After the same shock:

- ▶ AS shifts further to AS_3 (large shift)
- ▶ π^e rises ⇒ **second-round effects**
- ▶ Economy moves to point 3
- ▶ **Larger** decline in output
- ▶ **Larger** rise in inflation

Credibility and Aggregate Supply Shocks



With high credibility, AS shifts only to AS_2 (point 2). With low credibility, AS shifts further to AS_3 (point 3), causing more inflation and a deeper recession.

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Application: Oil Price Shocks — 1970s vs. 2007

1970s Oil Shocks

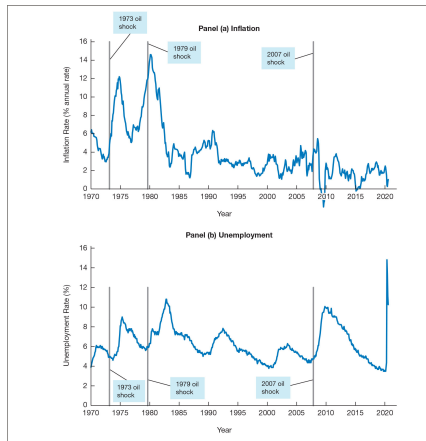
- ▶ Fed had **weak credibility**
- ▶ People expected inflation to **persist**
- ▶ π^e rose sharply $\Rightarrow$ AS shifted further
- ▶ Result: **stagflation** — high inflation *and* high unemployment

2007 Oil Price Shock

- ▶ Fed had **strong credibility** (built since Volcker era)
- ▶ Inflation expectations remained **well-anchored**
- ▶ AS shifted only modestly
- ▶ Result: **smaller** inflation increase, **milder** economic impact

$\Rightarrow$ The **same type of shock** produced vastly different outcomes because of differences in **credibility**.

Inflation and Unemployment, 1970–2020



The 1973 and 1979 oil shocks caused severe stagflation. The 2007 oil shock had a much smaller effect on inflation, reflecting improved Fed credibility.

Mishkin, The Economics of Money, Banking, and Financial Markets

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Credibility and the Cost of Disinflation

With Credibility

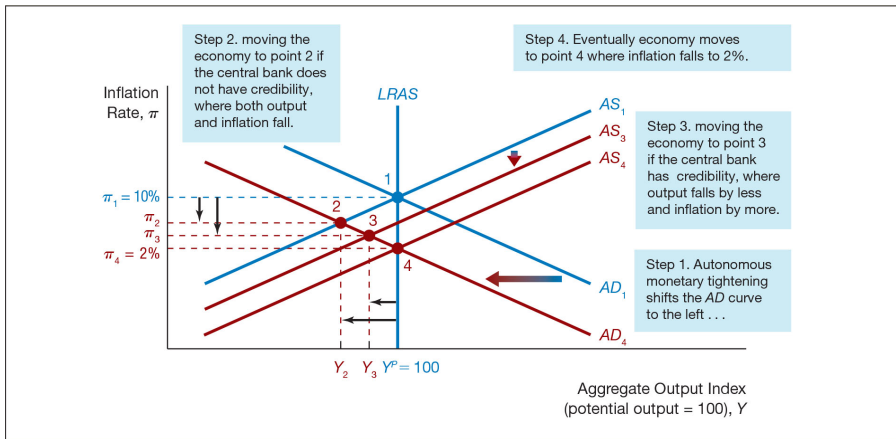
- ▶ Central bank announces it will **lower inflation**
- ▶ People **believe** the announcement
- ▶ π^e falls quickly
- ▶ AS shifts **down** $\Rightarrow$ inflation falls with **little output loss**
- ▶ Economy moves from point 1 to point 3

$\Rightarrow$ Credibility dramatically **reduces the sacrifice ratio** — the output cost per percentage point of disinflation.

Without Credibility

- ▶ Central bank announces disinflation
- ▶ People **don't believe** it
- ▶ π^e stays high $\Rightarrow$ AS does not shift
- ▶ The only way to reduce inflation is through a **deep recession**
- ▶ Economy moves from point 1 to point 2

Credibility and Anti-Inflation Policy



With credibility, AS shifts down quickly (point 3): inflation falls with less output loss. Without credibility, the economy must endure a deep recession (point 2) to achieve the same disinflation.

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Application: The Volcker Disinflation (1979–1983)

The Problem

By 1979, U.S. inflation had reached nearly **14%**. The Fed had **low credibility** after years of accommodating inflation.

The Policy

Fed Chair **Paul Volcker** raised the federal funds rate to over **20%**, deliberately inducing the worst recession since the Great Depression.

The Outcome

- ▶ Inflation fell from 14% to about **3%** by 1983
- ▶ Unemployment peaked at **10.8%** in late 1982
- ▶ The high cost reflected the Fed's **lack of initial credibility**
- ▶ But the Volcker disinflation **established credibility** that benefited the Fed for decades

Credibility as a Policy Multiplier

	High Credibility	Low Credibility
Demand shock impact	Small	Large
Supply shock impact	Contained	Amplified
Cost of disinflation	Low	High
Policy interventions needed	Small	Large
Expectations stability	Anchored	Volatile

The Virtuous Circle

Credibility $\Rightarrow$ anchored expectations $\Rightarrow$ smaller shocks $\Rightarrow$ better outcomes $\Rightarrow$ **more credibility**.

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How to Build Central Bank Credibility

Key Strategies

1. **Commit to a transparent inflation target**

- ▶ A public, numerical target gives the public a clear benchmark

2. **Maintain central bank independence**

- ▶ Insulate monetary policy from short-term political pressures
- ▶ Legal independence + operational independence

3. **Communicate clearly and frequently**

- ▶ Press conferences, minutes, forward guidance
- ▶ Explain the reasoning behind policy decisions

4. **Build a track record**

- ▶ Credibility is earned through **consistent actions over time**
- ▶ A single broken promise can destroy years of trust

Threats to Credibility

What Can Undermine Credibility?

1. Political pressure

- ▶ Governments may pressure central banks to finance deficits or stimulate before elections

2. Fiscal dominance

- ▶ If government debt is very high, the central bank may be forced to keep rates low to prevent a debt crisis
- ▶ Monetary policy becomes **subordinate** to fiscal needs

3. Inconsistent actions

- ▶ Announcing a 2% target but repeatedly tolerating 4–5% inflation
- ▶ “Actions speak louder than words”

4. Excessive focus on short-run output

- ▶ Sacrificing the inflation target to avoid short-term pain

Central Bank Independence: The Evidence

Countries with more **independent** central banks tend to have **lower and more stable inflation**, without sacrificing long-run economic growth.

Types of Independence

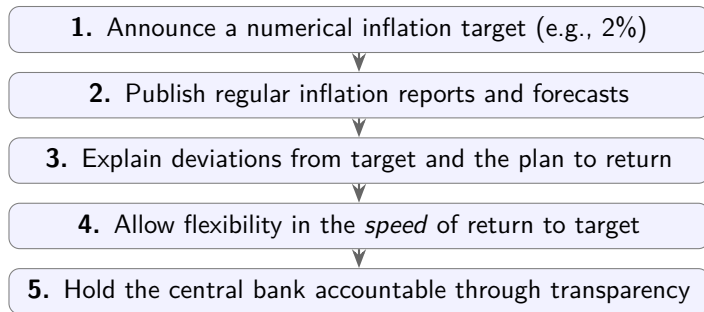
- ▶ **Goal independence**: the central bank sets its own objectives
- ▶ **Instrument independence**: the central bank chooses *how* to achieve the government's objectives
- ▶ Most modern central banks have **instrument independence** but not full goal independence

Examples

- ▶ **Bundesbank** (Germany, pre-euro): highly independent, delivered low inflation for decades
- ▶ **Bank of England**: granted operational independence in **1997**; inflation fell and stabilized

Modern Best Practices for Credibility

Most successful central banks today follow a version of **constrained discretion** through inflation targeting:



Chapter Summary

1. The **Lucas Critique**: policy evaluation using historical models is unreliable because expectations change when policy changes.
2. The **time-inconsistency problem** creates an inflationary bias under pure discretion, providing a strong argument for policy rules.
3. **Constrained discretion** (e.g., inflation targeting) combines the credibility of rules with the flexibility of discretion.
4. A credible **nominal anchor** stabilizes inflation expectations, reducing the impact of both demand and supply shocks.
5. **Credibility** dramatically lowers the cost of disinflation — a credible central bank can reduce inflation with far less output loss.
6. Credibility is built through **independence**, **transparency**, **consistent actions**, and a **track record** of keeping promises.

Key Equations to Remember

Expectations-Augmented Phillips Curve

$$\pi = \pi^e + \gamma(Y - Y^P) + \rho$$

Taylor Rule

$$i = \pi + r^* + a_\pi(\pi - \pi^T) + a_Y(Y - Y^P)$$

Expectations Hypothesis of the Term Structure

$$i_n = \frac{1}{n} \sum_{k=0}^{n-1} i_k^e + \ell_n$$

Long-term rates reflect expected future short-term rates plus a term premium.